n = 83: closure base D4 (K' = 4, 400 orbits) against M38 (K' = 16, 608 orbits)

Status (2026-11-08): result in. Verdict: improvement; every gate and control passes and the run is valid `[pcc:unverified]`. R = E2E(M38) / E2E(D4) = **2.0014 [1.8777, 2.1308]** over 60 alternating pairs, run 42425640847, Tier 1, instance scope. The design sections below are the frozen text, unchanged; they were frozen before any timing on the Tier 1 runner (commit `ebef3b260`). Every figure in them that comes from the shared cloud-agent VM is a local diagnostic, used only to size the design. The result section is added with the archived receipt.

Owner ruling, 2026-11-07 (recorded verbatim)

> Owner ruling, 2026-11-07, given in chat: a closure base with K' >= 4 > unknown-log columns counts as an index-calculus row. K' = 0 stays a > known-log reference row.

So D4 (K' = 4) is an index-calculus row, and this measurement is judged by the success criterion: a measured gain over the current n = 83 row. A K' = 0 base would be a known-log reference row (`koblitz-n83-knownlog-20261019`) and is not measured here. The ruling is also in `protocol.json` (`owner_ruling`) and in `docs/ic/BASELINES.md`, beside the known-log ruling-pending note.

Result

Run 42425640847 (workflow `n83-k4c400.yml`, push trigger, head `36866ad3c`, attempt 1, conclusion success) on one GitHub-hosted `ubuntu-24.04` runner (`GitHub Actions 1000197724`, image 20261101.360.1, **Intel Xeon Platinum 8370C**, execution CPU 0; the field-ops and field-ops-2 runs were on AMD EPYC 9V74). The schedule ran 50.7 minutes. The receipt is archived in `run_github_42425640847/` (`panel/runs.jsonl`, 2.4 MB, in the artifact store). The protocol and schedule are those of the frozen commit `ebef3b260`; commit `36866ad3c` changed only the workflow file (the artifact path and the store-pull step).

Verdict: improvement `[pcc:unverified]`. R = sum E2E(A) / sum E2E(B) = **2.0014 [1.8777, 2.1308]** over 60 alternating pairs (20 points x 3 pairs; two-stage paired bootstrap, 10,000 draws), A = M38 (K' = 16, 608 orbits), B = D4 (K' = 4, 400 orbits), the same binary `ic_online_fieldops_r770`. All 462 scheduled rows ran and are valid, none was skipped by the time guard, and every gate passes (`verify-stdout.json`): per base one index and layout digest set equal to its frozen one; Q0 equal to each base's frozen identity; identical relation digests, probe vectors, root calls and counters per D4 rank seed across kD4FO and kD4FOaa; rank tables equal to the frozen pre-checked tables; every IC row's pipeline names equal to its arm's; one scalar per point across both bases and rT, replayed in the oracle; placement, host identity and binaries unchanged. `verify_receipt.py` and `band_model.py` re-run on a copy of the receipt reproduce `panel/result.json`, `panel/result_table.md` and `band/band_model.json` byte for byte.

SVG).](figures/result.svg)

Primary, stages and controls (frozen rule)

stage (A = M38, B = D4)	side A mean (ms)	side B mean (ms)	A / B [95%]
index build (cold process)	5,116	2,138	2.3928 [2.3794, 2.4065]
scan self-test	1,093	220	4.9689 [4.8934, 5.0602]
rank stage (rank process)	3,815	2,182	1.7485 [1.5039, 2.0287]
online, one target	298	460	0.6487 [0.3347, 1.1395]
cold process	6,888	3,166	2.1757 [2.0530, 2.2975]
end to end (rank + cold)	10,703	5,348	**2.0014 [1.8777, 2.1308]**

The four stage rows are stage secondaries and never decide. The saving is in the setup (setup A / B 2.4353 [2.4198, 2.4516]: the index build over 2.4x fewer stored states, and the self-test 5.0x faster on the smaller Feistel domain) and in the rank stage (4 attempts instead of 16, each 2.4x more probes). The online stage is slower on D4 (A / B 0.6487, interval spanning 1: one target, one exponential sample per process). Process CPU time A / B 2.0715 [1.9855, 2.1626], process wall 2.0713; peak RSS 1,898 MB against 915 MB (A / B 2.0740) `[pcc:unverified]`.

- **A/A** (B against Baa, 40 pairs): end to end 1.0022 [0.9982, 1.0064] (interval contains 1), cold 1.0007 [0.9960, 1.0053] (inside [0.97, 1.03]), CPU time 1.0016 [0.9991, 1.0043].
- **Planted:** cD4FOX / cD4FO index build 2.6961 [2.6802, 2.7099] (lower bound above 1); the planted processes kept the frozen layout digests and checked 2 x stored states roots.
- **Probe law** (estimate, band [0.80, 1.20], both gated over at least 64 attempts): kD4FO **1.0999** over 240 attempts (bootstrap interval [0.9383, 1.2634]; its upper end is above the band, the gated estimate is inside), kM38FO 1.0018 [0.9233, 1.0897] over 960 attempts. The risk named before the run (the prototype's 1.154) did not materialise as a failure, but D4's point estimate is 10% above the law and its interval reaches 1.26: the true law ratio on D4 is not tight. kD4FO ns per probe 121.4.
- **Order and spread:** A first 2.0361, B first 1.9691; pair positions 2.0007, 2.0307, 1.9736; per-point ratios from 1.58 to 2.38 (20 of 20 above 1). Geometric mean of the per-pair ratios 2.0242 [1.9049, 2.1519].
- **Observed variance against the power check** (`observed_cv_per_side`, per pair): D4 rank stage 0.415 (modelled 0.50), online 0.79 (1.2), E2E 0.176 (0.23); M38 rank stage 0.222 (0.21), online 1.08 (1.2), E2E 0.090 (0.076). The model over-stated the D4 spread, so the design had at least the power `analysis/power.json` gives (0.998 at a true 1.20x).

The first run of the same frozen design (log only)

Run 42420993318 (head `ebef3b260`, the frozen commit) ran the same protocol and schedule and finished its whole schedule. Its job log printed the verifier's output: decision `improvement`, E2E(M38) / E2E(D4) = **1.964 [1.841, 2.091]**, every check true (including the probe law and the 64-attempt gate), A/A and planted controls passing. This is as reported from the job log by the coordinator `[pcc:unverified]`: its artifact step named the field-ops receipt directory, so **no receipt was uploaded and nothing from that run is archived or re-verified here**. It is reported beside the archived run, not combined with it. The two estimates (1.964 and 2.0014) have overlapping intervals; they are two draws of the same design (same points, same seeds, same schedule, two runner instances), so the run-to-run difference is the instance-to-instance noise of this measurement.

Distance to rho

In this run's own matched rT770 rho (20 processes, mean cold 2,610.5 ms, step 275.68 ns), the end-to-end cold gap is **4.0999 for side A (M38) and 2.0485 for side B (D4)** (point estimates, no interval computed for the gaps in this run); S end to end cold is 0.4194 for A and **0.2095 for B** in rT770 steps `[pcc:unverified]`. These are in-run gaps on an Intel Xeon Platinum 8370C. The field-ops run read 4.096 for the same M38 code on AMD EPYC 9V74 and the field-ops-2 run 3.165 for R1 on another panel and runner: gaps do not compose across runs (the matched rho's cold mean is a mean of random-length processes and moves with panel and host), only paired deltas within one run are deltas. Wall time, one host class, factor-base construction charged to neither side, so cold S is a lower bound.

Section 5a (priced native work; a model band)

`band_model.py` (frozen) with rho counted in this run and the IC arms modelled from the archived counted run 39882571652 (per-probe and per-attempt composition, this run's measured probes and attempts, each base's exact entries): at 0, 0.1 and 1 GAE per unpriced native operation, the gap to rT is A (M38) 0.000332, 1.136 [0.8876, 1.518], 4.527 and B (D4) **9.771e-05, 0.7272 [0.5657, 0.9728], 2.897**. **D4 is cheaper than M38 at every weight** (the order holds at both ends of every bound); **against the matched rho both are unresolved**: the order flips inside the band (D4 is below rho at w = 0 and w = 0.1, and above it at w = 1; M38 is below at w = 0 and above at w = 0.1 and 1). This is a derived model, not a counted run of the field-ops producer; the counts it uses are exact per (point, seed, base) so the band is the same in the first run; the linear algebra and factor-base construction are not charged and the intervals carry no model uncertainty `[pcc:unverified]`.

Limits of this result

- Wall time on one GitHub-hosted runner (Intel Xeon Platinum 8370C), one pinned CPU, Tier 1 instance scope, no host-wide exclusivity; not replayed on another host class (the field-ops and field-ops-2 runs were on AMD EPYC 9V74).
- **Not stacked on R1.** D4 was measured against the field-ops producer on the M38 base, the row that preceded field-ops-2 (R1, `koblitz-n83-fieldops2-20261107`, run 42424680756, 1.0256 [1.0236, 1.0278] over the same field-ops M38 reference). R1 x D4 was not run, and ratios from different runs do not compose.
- Per-base identities of D4 were frozen from local runs of the same producer (reproducibility, not independent proof); the independent checks are the pinned-oracle verification of header and table and the oracle replay of every recovered scalar.
- The narrowed panel scan, the reseed and the A/A rule change are disclosed above.
- Factor-base construction is charged to neither side, as in every n = 83 row.

Files (result)

`run_github_42425640847/` (receipt, `band/`), `figures/make_result_figure.py`, `figures/result.svg`, `figures/result.png`, `figures/make_report_pdf.py`, this report as `README.pdf`.

Question

The current n = 83 baseline row is the field-ops producer (`koblitz-n83-fieldops-20261104`, run 41869146827) on the M38 closure base: 16 unknown-log columns (K' = 16), 608 orbits in the root index. Its end-to-end cold cost splits (that run's side B, ms): index build 4,439, scan self-test 899, other setup 307, rank stage 3,160, online 171; E2E 8,976 `[pcc:unverified]`.

Is the end-to-end cold cost of the D4 closure base lower than the M38 row's, with the SAME frozen merged producer binary on both sides, on new held-out points and new rank seeds? Two levers move together (shortlist `algo_levers.md`, levers 1 and 2):

	M38 (row)	D4
`make_closure_base.py` flags	`--columns 16 --norm 64`	`--columns 4 --norm 165`
unknown-log columns K' = rank attempts	16	4
multipliers per column / orbits	38 / 608	100 / 400
logical states 83 K_orb^2	30,682,112	13,280,000
Feistel self-test domain	2^26	2^24 (logical states < 2^24 = 16,777,216)
stored (folded) states	15,341,360	6,640,200
table entries E	30,366,551	12,618,968
lost to within-column coincidences	1.03%	4.98%
table slots	2^26	2^25
probes per relation, law r / (166 E)	1.700 M	4.091 M

Fewer columns mean fewer relations (the rank stage needs K' of them); fewer orbits mean a smaller index to build and to self-test; the price is a smaller table, so each relation and each target costs 2.41 times more probes. `inputs/local_identity.json` holds the frozen D4 identity (digests, layout, Q0 identity), obtained from five local processes of the frozen producer; it is a reproducibility identity, not an independent proof. The independent checks of D4 are `verify_closure.py --base --table --planted` in the pinned pure-Python oracle (header, every orbit and label; table: every [L_c]G = R_c and [w L_c]G = P; three planted corruptions rejected), the table equal log for log to the first four logs of the oracle-checked M38 table, and, in the run, the oracle replay of every recovered scalar and the equality of every rank process's own produced table with the frozen one.

Local prototype (diagnostic)

A local prototype (loaded 4-core VM, one pinned core, 65 runs; `inputs/local_profile.json`, `inputs/local_prototype_rows.jsonl`, `profile/prototype_driver.py`) measured four candidate headers against M38. D4 is its header (d): mean process total over 10 rank seeds 5,221 ms (sd 595) against 9,669 ms for M38 (sd 879), a 1.85x local E2E ratio. That figure has no interval, one online target per cold run is a single exponential sample, the D4 rank stage is heavy-tailed, nothing is matched to rho, and the host was loaded: it is the reason to run the Tier 1 measurement, not a result `[pcc:unverified]`.

Design

One GitHub-hosted `ubuntu-24.04` runner is the Tier 1 instance; one pinned CPU, one fresh process per run (`run_schedule.py`, the template's, byte for byte).

- **One binary on both sides:** `ic_online_fieldops_r770`, the field-ops crate (`../koblitz-n83-fieldops-20261104/online-b`) built unchanged with `-C target-feature=+pclmulqdq`; every environment variable, pipeline name and gate of an IC process is the field-ops experiment's. Only the base header and its factor-log table differ (`arm.py` gates every IC process against ITS arm's base: base hash, columns, orbits, orbits per column, the multiplier list, states, slots, entries, digests, layout digests, and the cached table's own digest). `restore.py` runs the field-ops `restore.py` with its pin check (every digest of that protocol), regenerates the M38 and D4 headers to their frozen SHA-256 and pins the field-ops files this experiment builds and runs.
- **Panel:** 20 new held-out points (`inputs/cold_targets.json`, role `k4c400-cold-v1`, disjoint from every earlier n = 83 panel including the field-ops and field-ops-2 panels: `inputs/panel_disjointness.json`).
- **Pairs per point:** three alternating primary pairs (A = M38 side, B = D4 side, same point and same rank seed, ABAB or BABA with a seeded start), two A/A pairs (B against Baa, the D4 arms twice), one planted pair (cD4FO against cD4FOX, `KIC_INDEX_CANON=lift_plant`) and one tuned-rho process.
- **Rank seeds:** 2501-2560, 60 distinct seeds (`inputs/seed_disjointness.json`; the first choice, 2301-2360, collided with `koblitz-n83-onlineframe2-20261021/schedule.json` on main and three branches, was caught by `check_seed_disjoint.py` (`new: false`) and replaced before any timing): 240 D4 attempts and 960 M38 attempts for the probe-law gates.
- **Primary:** R = sum E2E(A) / sum E2E(B) over the 60 primary pairs, E2E = rank stage + cold process. Decision: improvement iff the 95% two-stage paired bootstrap lower bound of R exceeds 1 (10,000 draws, points then pairs, A and B of a pair drawn together) `[pcc:unverified]`.
- **Per stage** (stage secondaries, never deciding): index build, scan self-test, rank stage, online.
- **Q0:** one cold process of cM38FO and cD4FO on the validation point; each must reproduce its base's frozen identity.

Arms: side A `kM38FO` + `cM38FO` (verbatim from the field-ops protocol); side B `kD4FO` + `cD4FO`; A/A `kD4FOaa` + `cD4FOaa`; planted `cD4FO` / `cD4FOX`; matched rho `rT` (the tuned rT770).

Gates

The field-ops verifier's gates with these changes (`verify_receipt.py`): two bases, each against its own frozen identity (the digest sets must be exactly {M38, D4}); rank-seed identity (relation digest, probe vector, root calls, counters) across kD4FO and kD4FOaa; **probe law for both kD4FO and kM38FO inside [0.80, 1.20] over at least 64 attempts each** (a new gate: 240 and 960 attempts are scheduled); every scalar replayed in the oracle; placement, host identity, binary identity, pipeline names, rank tables equal to the frozen pre-checked tables.

Controls

- **A/A:** B against Baa. The cold ratio's point estimate must be inside [0.97, 1.03]; the end-to-end ratio's 95% interval must contain 1. The field-ops template required the end-to-end *point estimate* inside [0.97, 1.03]; with independent rank and online draws on D4 that rule fails a true null about half the time (`analysis/power.json`: `e2e_point_inside_0.97_1.03_rate` 0.47), so it is replaced here by the interval.
- **Planted:** cD4FOX / cD4FO index build, lower bound above 1, layout digests unchanged, 2 x stored states roots checked.
- **Exactness before timing:** the planted corruptions rejected on both closure bases, the crate tests.

Power and variance (a sizing simulation, not a result)

`power_check.py` draws simulated receipts from the field-ops archive (the M38 side) and models D4 (index build x 0.399 from the local cold ratio, self-test x 0.25 from the domain step, online and each of 4 rank attempts x 2.406 from E_M38 / E_D4, ns per probe unchanged; the rank stage is a sum of 4 archived attempt times, so its heavy tail is kept). Per-pair coefficients of variation in the model: D4 rank stage 0.50, online 1.2, E2E 0.23 (M38: rank 0.21, online 1.2, E2E 0.076). Because A and B run different bases their noise is independent, so the design resolves a large effect, not a small one:

true E2E ratio	1.00	1.05	1.10	1.20	1.50	modelled (1.95)
power, 12 x 3 pairs	0.03	0.15	0.47	0.94	1.00	1.00
power, 16 x 3 pairs	0.02	0.16	0.53	0.98	1.00	1.00
power, 20 x 3 pairs (frozen)	0.01	0.21	0.70	1.00	1.00	1.00

The frozen 20 x 3 design separates an E2E ratio of 1.2 or more with power above 0.99, and will not separate 1.05 (0.21) or reliably 1.10 (0.70): a smaller true effect is reported as not separated, not as a regression. More seeds help the heavy-tailed rank stage but the 60 seeds also fix the probe-law gate: at a true law ratio of 1.00 it passes with probability 0.998; at 1.15 (the local prototype's 40-attempt point estimate) 0.83. The modelled schedule is about 43 min (job about 62 min; about 92 min on a 1.5x slower host; guard 165 min).

Section 5a (priced native work against the matched rho)

AGENTS.md section 5a prices every arm's unpriced native work, the matched rho's included, at 0, 0.1 and 1 GAE per operation and ranks two arms only where the order holds across that band. `band_model.py` runs the opcounts experiment's `band.py` unchanged. **rho is counted in this run** (every rT process's own `charges_setup_and_solve`). **The field-ops producer emits no operation counts**, so the IC arms are a **model**: the archived counted run 39882571652's per-probe and per-attempt composition, applied to this run's measured probes and attempts per side and point, and to each base's exact stored entries. It is a derived model band, labelled model wherever it is quoted, not a counted run of this producer; the linear algebra and factor-base construction are not charged and the intervals carry no model uncertainty.

Disclosures made before timing

- **Narrowed panel scan.** The field-ops `check_panel_disjoint.py` scans every origin branch's changed files. Here that was about 1,270 branches and did not finish in a 30-minute background limit. This copy scans origin/main in full and only the 90 origin branches whose name contains `n83` (2,808,268 files; 0 hits), plus the local worktree's `experiments/` (65,679 files; 0 hits); `refs_filter` in `inputs/panel_disjointness.json` says so. `make_targets.py` also regenerates every known role of the law and found no shared point. The scanner is no longer byte-identical to the field-ops file and is pinned separately.
- **Reseed.** Rank seeds 2301-2360 collided with `koblitz-n83-onlineframe2-20261021/schedule.json`; 2501-2560 are used.
- **A/A rule.** The end-to-end A/A rule is the interval containing 1 (above); the template's point-estimate band is kept for the cold ratio.
- **Local smoke** of segments O1 and P1 through `run_schedule.py` and `arm.py` (25 rows, all valid, every correctness check true except the 64-attempt law gate that a partial receipt cannot meet): a correctness check, never timing evidence.

Not in scope

The producer, the probe loops, the rank-stage algorithm, the self-test's predicate and the factor base are unchanged. Headers a (K' = 8, 624 orbits), b (K' = 4, 612) and c (K' = 8, 448) of the prototype are not measured here. Factor-base construction is charged to neither side, as in every n = 83 row.

Trigger

`.github/workflows/n83-k4c400.yml`, on a push to `claude/n83-k4c400` restricted by path filters (as the field-ops workflow; `workflow_dispatch` was refused for earlier n = 83 integrations). It clones crypto 770f6a69 from the public `isogeny-labs/crypto`.

Files

`restore.py`, `make_log_table.py`, `local_identity.py`, `make_local_profile.py`, `make_targets.py`, `check_panel_disjoint.py`, `check_seed_disjoint.py`, `make_power_prior.py`, `power_check.py`, `make_protocol.py`, `make_schedule.py`, `run_schedule.py`, `arm.py`, `check_log_table.py`, `verify_receipt.py`, `test_verify_receipt.py`, `band_model.py`; `protocol.json`, `schedule.json`; `inputs/`, `analysis/power.json`, `profile/`.